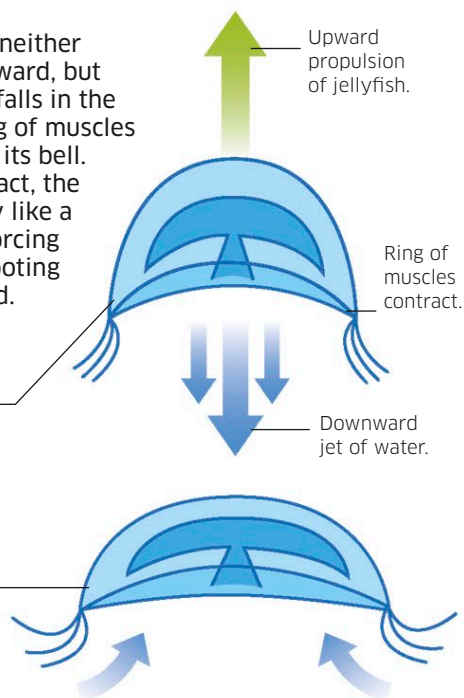


Pulsating

A jellyfish moves neither forward nor backward, but instead rises and falls in the water. It has a ring of muscles around the rim of its bell. When these contract, the bell closes slightly like a drawstring bag, forcing water out and shooting the animal upward.

Power stroke
Muscles contract to propel the jellyfish upward.

Recovery stroke
Water fills the bell as muscles relax before the power stroke repeats.



Getting around

Whether over land, underwater, or in the air, animals can move themselves around in extraordinary ways when all their muscles work together.

Although all living things move parts of their body to an extent, only animals can truly “locomote”. This is when the entire body moves to a different location. Some animals do it without any muscle power at all—riding on ocean currents or getting blown by the wind. But most animals locomote under their own steam. They do so for many different reasons: to find food or a mate, or to escape from predators. Some animals migrate over enormous distances from season to season, or even from day to day.

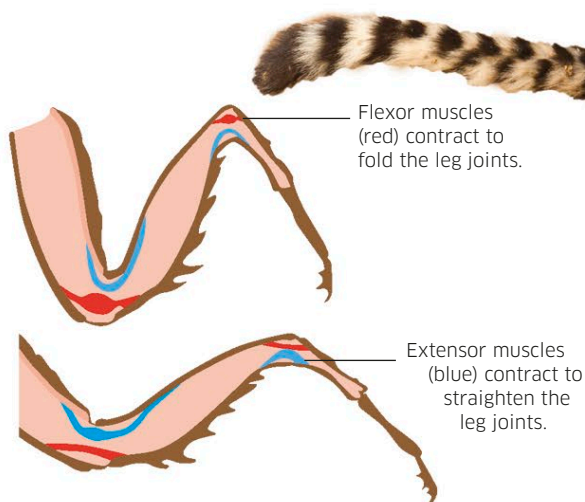
Tiny, deep ocean pygmy sharks grow no bigger than 8 in (20 cm), but each night swim 1 mile (1.5 km) up to the surface and back in order to feed.



Tiger beetles have long legs to give them speed.

Tiger beetle

Predatory tiger beetles are fast sprinters. Like all insects, they have six legs, and when running they lift three simultaneously, leaving three in contact with the ground. However, their big eyes cannot keep up with their speed, meaning their vision is blurred every time they run.

**Multi-jointed leg**

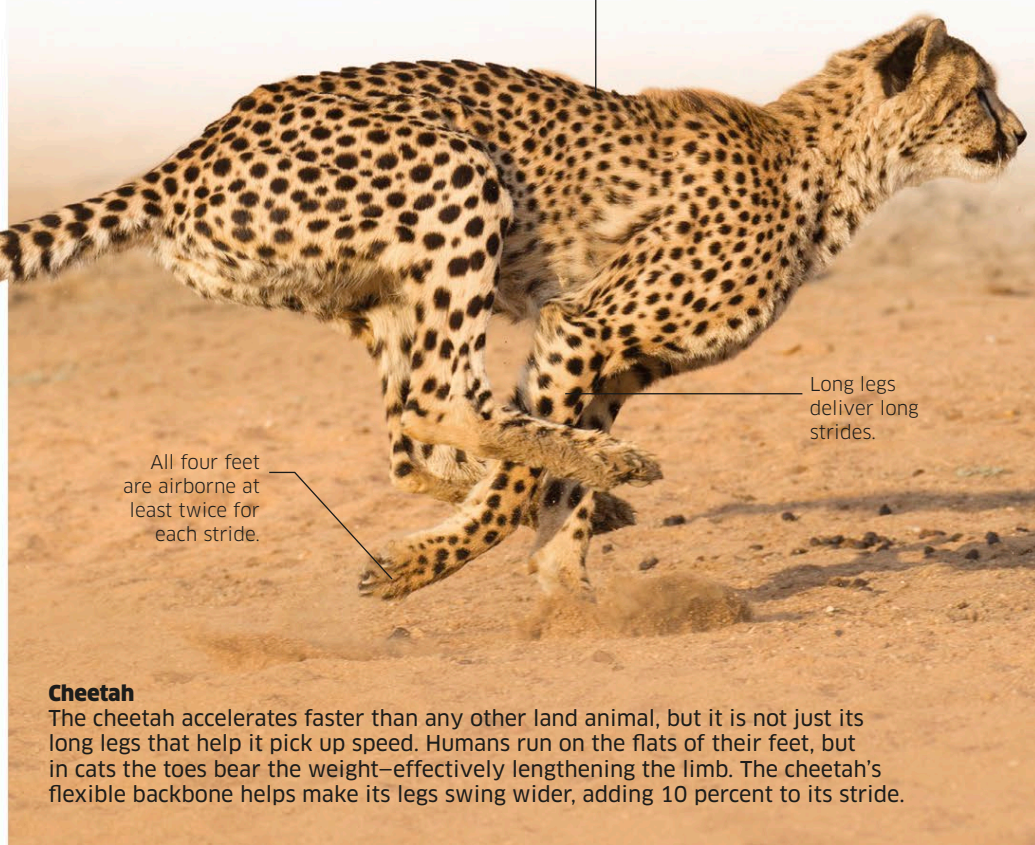
Arthropods, including insects, spiders, and crabs, have multi-jointed legs that carry an armorlike outer skeleton, with their muscles attached on the inside. Their muscles work in pairs around each joint—one to flex (bend) and the other to extend.

Running

An animal that moves over land needs its muscles to pull against a strong supporting framework. It also needs good balance to stay upright, meaning that its muscles and skeleton must work together with the nervous system. Some animals move slowly, even when in a hurry, but others are born to run. The fastest runners not only have powerful muscles to move their limbs more quickly, but also take much longer strides.

A cheetah can accelerate to 62 mph (100 km/h) in just three seconds.

A flexible spine helps the cheetah's body bend up and down when sprinting.



All four feet are airborne at least twice for each stride.

Long legs deliver long strides.

Cheetah

The cheetah accelerates faster than any other land animal, but it is not just its long legs that help it pick up speed. Humans run on the flats of their feet, but in cats the toes bear the weight—effectively lengthening the limb. The cheetah's flexible backbone helps make its legs swing wider, adding 10 percent to its stride.